

18.4 Randomized complete blocks

A covariate removes nuisance variation by regression. **Blocking** removes it by design: similar units (plots in one part of a field, animals from one litter, strips cut from one board) are grouped into blocks, and treatments are compared *within* blocks. This section analyses the randomized complete block design, explains how randomization justifies its model, and measures what blocking gains.

18.4.1 The design and its model

Definition 18.4.1 — Randomized complete block design

There are t treatments and b blocks of t units each. In every block, the t treatments are assigned to the t units by a random permutation, chosen uniformly and independently for different blocks. The response of the unit in block j that received treatment i is written y_{ij} .

The usual model is the additive two-way model with one observation per cell,

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad (18.4.1)$$

with treatment effects τ_i and block effects β_j . Its geometry is Example (Additive two-factor model, one observation per cell) and its analysis of variance is that of Theorem 16.1.2; we restate it in the form the design needs.

Theorem 18.4.1 — Analysis of the randomized complete block design

Assume Eq. 18.4.1 with $t, b \geq 2$. Let $\bar{y}_{i\cdot}$, $\bar{y}_{\cdot j}$ and $\bar{y}_{\cdot\cdot}$ be the treatment, block and grand means.

- (a) The sums of squares

$$\begin{aligned} SS_T &= b \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2, & SS_B &= t \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2, \\ SSE &= \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \end{aligned}$$

are squared lengths of projections onto mutually orthogonal subspaces of dimensions $t-1$, $b-1$ and $(t-1)(b-1)$, and with the correction for the mean they add up to $\sum_{i,j} y_{ij}^2$.

- (b) A function $\sum_i c_i \tau_i$ is estimable iff $\sum_i c_i = 0$. Its least squares estimate is $\sum_i c_i \bar{y}_{i\cdot}$, and

$$\sum_i c_i \bar{y}_{i\cdot} = \sum_i c_i \tau_i + \sum_i c_i \bar{\varepsilon}_{i\cdot}$$

identically, whatever the block effects. Its variance is $\sigma^2 \sum_i c_i^2 / b$.

(c) The treatment F statistic satisfies

$$F_T = \frac{SS_T/(t-1)}{SSE/((t-1)(b-1))} \sim F(t-1, (t-1)(b-1), \gamma),$$

with

$$\gamma = \frac{b}{\sigma^2} \sum_i (\tau_i - \bar{\tau})^2.$$

(d) $E(SS_B/(b-1)) = \sigma^2 + t \sum_j (\beta_j - \bar{\beta})^2/(b-1)$, and $E(SSE/((t-1)(b-1))) = \sigma^2$.

Proof. Model [Eq. 18.4.1](#) is the additive two-way model of Theorem 16.1.2 with factor A the treatments, factor B the blocks, $a = t$ and $m = 1$, so that $n - a - b + 1 = (t-1)(b-1)$. Parts (a), (b) and (c) are Theorem 16.1.2(a), (b) and (c), with (e) for the form of the residual space. The only new statement is the identity in (b): averaging [Eq. 18.4.1](#) over j gives $\bar{y}_{i.} = \mu + \tau_i + \bar{\beta} + \bar{\varepsilon}_{i.}$, and $\sum_i c_i(\mu + \bar{\beta}) = 0$ for a contrast. Part (d) is Theorem 9.6.1 applied to the decomposition in (a); the block projection sends the mean vector to the vector with entries $\beta_j - \bar{\beta}$, whose squared length is $t \sum_j (\beta_j - \bar{\beta})^2$. ■

Part (b) is the reason to block. However large the block effects, they cancel exactly from every treatment comparison and from the error mean square, which estimates the variation *within* blocks. The between-block variation moves into its own line, SS_B . A test of blocks is rarely of interest, since blocks were chosen because they differ; a large MS_B measures what blocking achieved ([Proposition 18.4.4](#) below).

Any block-by-treatment interaction is absorbed into the error, and without replication within cells it cannot be separated from it. Tukey's one-degree-of-freedom test (Theorem 16.1.2(f)) checks for its most common form. If treatment differences genuinely vary between blocks, the comparisons refer to the average over the blocks used.

18.4.2 Randomization as the justification of the model

The units of an experiment are not a sample from a normal population; they are the boards or patients at hand. What the experimenter controls is the randomization, and it alone supports much of the analysis. Give each unit u a fixed value ξ_u , and assume **unit–treatment additivity**: unit u responds to treatment i with $\xi_u + \tau_i$. The only probability below is that of the randomization.

Proposition 18.4.2 — Randomization moments of the block analysis

Let the units of block j have values $\xi_{j1}, \dots, \xi_{jt}$, with mean $\bar{\xi}_j$ and $W_j = \sum_s (\xi_{js} - \bar{\xi}_j)^2$, and put

$$S_W^2 = \frac{1}{b(t-1)} \sum_j W_j,$$

the average within-block variance of the unit values. Under unit–treatment additivity

and the randomization of [Definition 18.4.1](#):

- (a) every contrast estimate $\sum_i c_i \bar{y}_i$ has randomization mean $\sum_i c_i \tau_i$ and randomization variance $S_W^2 \sum_i c_i^2 / b$;
- (b) $E(\text{SSE} / ((t-1)(b-1))) = S_W^2$;
- (c) $E(\text{SS}_T / (t-1)) = S_W^2 + b \sum_i (\tau_i - \bar{\tau})^2 / (t-1)$.

So the error mean square is unbiased for the variance that the model attributes to σ^2 , and when the treatments have no effect the treatment and error mean squares have the same expectation.

Proof. Let $\pi_j(i)$ be the unit of block j that receives treatment i , and $w_{ij} = \xi_{j\pi_j(i)} - \bar{\xi}_j$. Then $y_{ij} = \bar{\xi}_j + \tau_i + w_{ij}$. For fixed j , the vector $(w_{1j}, \dots, w_{tj})$ is a uniformly random permutation of the deviations $\xi_{js} - \bar{\xi}_j$, which sum to zero. Hence $E w_{ij} = 0$, $E w_{ij}^2 = W_j / t$, and, since $\sum_k w_{kj} = 0$, $E w_{ij} w_{kj} = -W_j / \{t(t-1)\}$ for $i \neq k$. Different blocks are independent.

- (a) $\sum_i c_i \bar{y}_i = \sum_i c_i \tau_i + b^{-1} \sum_j \sum_i c_i w_{ij}$, because $\sum_i c_i = 0$ removes the $\bar{\xi}_j$. The mean follows. For the variance,

$$\text{Var}\left(\sum_i c_i w_{ij}\right) = \frac{W_j}{t} \sum_i c_i^2 - \frac{W_j}{t(t-1)} \sum_{i \neq k} c_i c_k = \frac{W_j}{t-1} \sum_i c_i^2,$$

since $\sum_{i \neq k} c_i c_k = (\sum_i c_i)^2 - \sum_i c_i^2 = -\sum_i c_i^2$. Summing over the independent blocks and dividing by b^2 gives $S_W^2 \sum_i c_i^2 / b$.

- (b) The block means are $\bar{y}_{\cdot j} = \bar{\xi}_j + \bar{\tau}$, and the treatment means are $\bar{y}_i = \bar{\xi} + \tau_i + \bar{w}_{i\cdot}$, with $\bar{\xi} = b^{-1} \sum_j \bar{\xi}_j$. So the residual is $w_{ij} - \bar{w}_{i\cdot}$, and

$$\text{SSE} = \sum_{i,j} w_{ij}^2 - b \sum_i \bar{w}_{i\cdot}^2.$$

The first sum is $\sum_j W_j$ for every randomization. By independence across blocks, $E \bar{w}_{i\cdot}^2 = b^{-2} \sum_j E w_{ij}^2 = b^{-2} \sum_j W_j / t$, so $E(b \sum_i \bar{w}_{i\cdot}^2) = b^{-1} \sum_j W_j$. Hence $E(\text{SSE}) = (1 - 1/b) \sum_j W_j = (t-1)(b-1)S_W^2$.

- (c) $\text{SS}_T = b \sum_i (\tau_i - \bar{\tau} + \bar{w}_{i\cdot})^2$, using $\sum_i \bar{w}_{i\cdot} = 0$. The cross terms have mean zero, and $E(b \sum_i \bar{w}_{i\cdot}^2) = (t-1)S_W^2$ by the computation in (b).

■

Independent measurement errors of variance σ_e^2 add σ_e^2 to every expectation. The proposition does not make F_T exactly F -distributed, but it shows that the estimates and their estimated variances are unbiased because of the randomization, whatever made the units differ. A systematic assignment could line treatments up with the ξ 's, and then neither statement would hold.

Randomization tests. Under the *sharp* null hypothesis that every unit would respond the same under every treatment, the responses are fixed and only the labels are random. The proportion of the $(t!)^b$ equally likely randomizations giving an F_T at least as large as the observed one is an exact p -value, estimated by sampling randomizations when there are too many. [Chapter 23](#) develops permutation tests for linear models.

18.4.3 An example

Example. (Five wood adhesives on six boards)

In a synthetic experiment, five adhesives are compared for the shear strength of glued joints. Timber varies from board to board, so each of six boards is cut into five strips, assigned to the adhesives by a random permutation. The strengths (MPa) are:

adhesive	board 1	board 2	board 3	board 4	board 5	board 6
1	7.89	8.42	8.55	8.30	9.26	9.15
2	8.68	8.69	8.73	8.62	9.72	9.51
3	9.13	9.28	9.81	9.43	10.30	9.41
4	8.48	8.60	8.46	8.95	9.42	9.02
5	8.90	9.17	9.79	9.57	10.24	9.09

The analysis of variance is:

source	df	sum of squares	mean square	F
adhesives	4	4.106	1.0266	14.88
boards	5	4.068	0.8136	11.79
error	20	1.380	0.0690	
corrected total	29	9.555		

The adhesives differ clearly ($p = 8.6 \times 10^{-6}$). The adhesive means are 8.595, 8.992, 9.560, 8.822 and 9.460, and every difference has standard error $\sqrt{2 \times 0.0690/6} = 0.152$. Tukey's honestly significant difference at level 0.05, $q_{0.05}(5, 20)\sqrt{0.0690/6}$, is 0.454 (Theorem 13.3.2 applies, with the error mean square of the block analysis, as in Exercise (B1)). It separates adhesives 3 and 5 from adhesives 1, 2 and 4: the significant pairs are 1–3, 1–5, 2–3, 2–5, 3–4 and 4–5. Tukey's test for nonadditivity gives $F = 0.07$ ($p = 0.80$), so there is no sign that the adhesive differences depend on the strength of the board.

Of 20000 randomizations sampled from the 2.99×10^{12} possible plans, 1 gave an adhesive F at least as large as 14.88: the randomization p -value is $(1 + 1)/(20000 + 1) = 0.00010$, in line with the normal theory.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(606)
t, b = 5, 6                                # adhesives, boards
plan = np.array([rng.permutation(t) for _ in range(b)]) # which strip of each
board gets which adhesive
tau = np.array([0.0, 0.40, 0.65, 0.15, 0.80]) # adhesive effects (unknown in
practice)
board = rng.normal(0, 0.75, b)               # board effects
y = np.round(9 + tau[:, None] + board[None, :] + rng.normal(0, 0.3, (t, b)), 2)

grand = y.mean()
trt_mean, blk_mean = y.mean(axis=1), y.mean(axis=0)
```

```

resid = y - trt_mean[:, None] - blk_mean[None, :] + grand
ss_trt = b * np.sum((trt_mean - grand) ** 2)
ss_blk = t * np.sum((blk_mean - grand) ** 2)
ss_err = np.sum(resid ** 2)
df_err = (t - 1) * (b - 1)
ms_trt, ms_blk, ms_err = ss_trt / (t - 1), ss_blk / (b - 1), ss_err / df_err
F_trt = ms_trt / ms_err
print(f"treatments SS {ss_trt:.2f}, blocks SS {ss_blk:.2f}, error SS {ss_err:.2f} on
      {df_err} df")
print(f"F for adhesives = {F_trt:.2f}, p = {stats.f.sf(F_trt, t - 1, df_err):.2g}")
print("adhesive means", trt_mean.round(2), " se of a difference", np.sqrt(2 * ms_err
      / b).round(3))

```

[Run this code online](#)

```

def f_stat(yy):
    tm, bm, gm = yy.mean(axis=1), yy.mean(axis=0), yy.mean()
    sst = b * np.sum((tm - gm) ** 2)
    sse = np.sum((yy - tm[:, None] - bm[None, :] + gm) ** 2)
    return (sst / (t - 1)) / (sse / df_err)

rng_perm = np.random.default_rng(7)
draws = 20_000
F_perm = np.empty(draws)
for r in range(draws):
    # re-randomize: permute the adhesive labels independently within every board
    perm = np.array([rng_perm.permutation(t) for _ in range(b)]).T
    F_perm[r] = f_stat(np.take_along_axis(y, perm, axis=0))
p_rand = (1 + np.sum(F_perm >= F_trt)) / (draws + 1)
print(f"randomization p-value {p_rand:.4f} (F-test p-value {stats.f.sf(F_trt, t -
      1, df_err):.2g})")

```

[Run this code online](#)

18.4.4 How much did blocking gain?

Had the adhesives been assigned to the thirty strips completely at random, the completely randomized design (CRD) would have had the board-to-board variation in its error. The block experiment itself can estimate how large that error would have been, using a fact about sampling from a finite population.

Lemma 18.4.3 — Sample variance without replacement

Let $u_1, \dots, u_N$ be fixed numbers with mean $\bar{u}$ and $S^2 = \sum_a (u_a - \bar{u})^2 / (N - 1)$. If $\mathcal{S}$ is a simple random sample of $m \geq 2$ of the indices, drawn without replacement, then

$$E \sum_{a \in \mathcal{S}} (u_a - \bar{u}_{\mathcal{S}})^2 = (m - 1)S^2,$$

where $\bar{u}_{\mathcal{S}}$ is the sample mean.

Proof. For any m numbers, $\sum_a (u_a - \bar{u}_S)^2 = \frac{1}{2m} \sum_{a,c} (u_a - u_c)^2$, the sum being over ordered pairs of sample members. The $m(m-1)$ ordered pairs with $a \neq c$ are each a uniformly random ordered pair of distinct population members. For such a pair, $E(u_a - u_c)^2 = \frac{1}{N(N-1)} \sum_{a \neq c} (u_a - u_c)^2 = \frac{2N \sum_a (u_a - \bar{u})^2}{N(N-1)} = 2S^2$. So the expectation is $\frac{1}{2m} m(m-1) 2S^2 = (m-1)S^2$. ■

Proposition 18.4.4 — What a completely randomized design would have given

Let the $N = bt$ units follow Eq. 18.4.1, with fixed block effects β_j , and write $S_\beta^2 = t \sum_j (\beta_j - \bar{\beta})^2 / (N-1)$ for the variance of the block effects over the N units.

- (a) If instead the treatments were assigned to the same units completely at random, b units each, the within-treatment mean square of the resulting one-way analysis would have expectation $\sigma^2 + S_\beta^2$, over the errors and the randomization.
- (b) In the block design,

$$\hat{\sigma}_{\text{CR}}^2 = \frac{(b-1)\text{MS}_B + b(t-1)\text{MSE}}{bt-1} \quad (18.4.2)$$

is unbiased for $\sigma^2 + S_\beta^2$.

Proof.

- (a) The b units given treatment i form a simple random sample of the N units. Their responses are $\mu + \tau_i + \beta_{j(u)} + \varepsilon_u$, so the within-treatment sum of squares of treatment i is that of the numbers $\beta_{j(u)} + \varepsilon_u$ over the sample. Given the assignment, the errors contribute $(b-1)\sigma^2$ in expectation, and the cross terms have mean zero. By Lemma 18.4.3 the block effects contribute $(b-1)S_\beta^2$ in expectation over the assignment. Summing over the t treatments and dividing by $t(b-1)$ gives $\sigma^2 + S_\beta^2$.
- (b) By Theorem 18.4.1(d), $E[(b-1)\text{MS}_B] = (b-1)\sigma^2 + t \sum_j (\beta_j - \bar{\beta})^2$ and $E[b(t-1)\text{MSE}] = b(t-1)\sigma^2$. The sum has expectation $(bt-1)\sigma^2 + (N-1)S_\beta^2$. ■

The **relative efficiency** $\hat{\sigma}_{\text{CR}}^2 / \text{MSE}$ estimates how many units a CRD would need per unit of the block design for the same precision. Some authors multiply it by Fisher's factor $(\nu_B + 1)(\nu_C + 3) / \{(\nu_B + 3)(\nu_C + 1)\}$, with $\nu_B = (t-1)(b-1)$ and $\nu_C = t(b-1)$ the two error degrees of freedom, to allow for estimating σ^2 with fewer degrees of freedom; this is a convention, not a theorem.

For the adhesives,

$$\hat{\sigma}_{\text{CR}}^2 = \frac{5 \times 0.8136 + 24 \times 0.0690}{29} = 0.1974,$$

and the relative efficiency is $0.1974/0.0690 = 2.86$ (2.81 with Fisher's factor). Blocking was worth almost three times as many strips; without it the adhesive F would have been about $1.0266/0.1974 = 5.20$.

```

s2_crd = ((b - 1) * ms_blk + b * (t - 1) * ms_err) / (b * t - 1)  # estimated CRD
                        error variance
re = s2_crd / ms_err
nu_b, nu_c = df_err, b * t - t                                     # error df: blocked,
                        completely randomized
re_fisher = re * (nu_b + 1) * (nu_c + 3) / ((nu_b + 3) * (nu_c + 1))
F_crd = ms_trt / s2_crd                                           # what the treatment F would
                        roughly have been
print(f"estimated CRD error variance {s2_crd:.2f} against {ms_err:.2f}:"
      f" relative efficiency {re:.2f} ({re_fisher:.2f} with Fisher's df factor)")

```

[Run this code online](#)

Remark (Fixed or random blocks). Treating the boards as a random sample of boards changes nothing about treatment comparisons in a complete block design, since block effects cancel from every contrast ([Theorem 18.4.1\(b\)](#)). It matters for incomplete blocks ([Section 18.5](#)) and for predicting a new block; Chapter 32 treats such mixed models.

18.4.5 Exercises

18.4.5.1 A. Check your understanding

Exercise A1.

An experiment has $t = 4$ treatments in $b = 8$ blocks. Give the degrees of freedom of the block design's error, and of the error of a completely randomized design with the same 32 units. How large must the relative efficiency (without Fisher's factor) be for blocking to be worthwhile?

Exercise A2.

Verify from the table in [Example \(Five wood adhesives on six boards\)](#) that the treatment mean of adhesive 3 is 9.560, and compute the difference between adhesives 3 and 5 with its standard error. Is it significant by Tukey's criterion?

18.4.5.2 B. Practice

Exercise B1.

The strips cut from a board differ systematically: strips from the outer edge are stronger. How would you modify the design and analysis to remove this source of variation as well? Which design of the next section does your answer lead to?

Exercise B2.

In [Proposition 18.4.2](#), show that $\text{Cov}(\sum_i c_i \bar{y}_{i.}, \sum_i d_i \bar{y}_{i.}) = S_W^2 \sum_i c_i d_i / b$ for two contrasts. Conclude that orthogonal contrasts are uncorrelated over the randomization, as they are under the normal model.

Solution. As in the proof, $\text{Cov}(\sum_i c_i w_{ij}, \sum_k d_k w_{kj}) = \frac{W_j}{t} \sum_i c_i d_i - \frac{W_j}{t(t-1)} \sum_{i \neq k} c_i d_k$, and $\sum_{i \neq k} c_i d_k = (\sum_i c_i)(\sum_k d_k) - \sum_i c_i d_i = -\sum_i c_i d_i$. So the covariance is $W_j \sum_i c_i d_i / (t-1)$. Summing over blocks and dividing by b^2 gives $S_W^2 \sum_i c_i d_i / b$, which is zero for orthogonal contrasts.

Exercise B3.

Suppose the block effects are random, $\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$, independent of the errors. Show that $E(\hat{\sigma}_{\text{CR}}^2) = \sigma^2 + \sigma_\beta^2 t(b-1)/(bt-1)$, and interpret the factor $t(b-1)/(bt-1)$.

18.4.5.3 C. Going deeper

Exercise C1.

In Eq. 18.4.1 let the block effects be random, $\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$, independent of the errors. Show that the treatment means $\bar{y}_{i\cdot}$ are equicorrelated, with variance $(\sigma^2 + \sigma_\beta^2)/b$ and covariance σ_β^2/b , while every contrast estimate keeps the distribution given by Theorem 18.4.1(b). Show that $E(\text{MS}_B) = \sigma^2 + t\sigma_\beta^2$, and that $(\text{MS}_B + (t-1)\text{MSE})/(tb)$ is unbiased for $\text{Var}(\bar{y}_{i\cdot})$. Why is $s/\sqrt{b}$ the right scale for comparing treatments but the wrong standard error for a single treatment mean?

Solution. Averaging over blocks, $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\beta} + \bar{\varepsilon}_{i\cdot}$, where $\bar{\beta} \sim N(0, \sigma_\beta^2/b)$ is common to all treatments and the $\bar{\varepsilon}_{i\cdot}$ are independent $N(0, \sigma^2/b)$ and independent of $\bar{\beta}$. Hence $\text{Var}(\bar{y}_{i\cdot}) = (\sigma^2 + \sigma_\beta^2)/b$ and $\text{Cov}(\bar{y}_{i\cdot}, \bar{y}_{k\cdot}) = \text{Var}(\bar{\beta}) = \sigma_\beta^2/b$ for $i \neq k$. In a contrast β is multiplied by $\sum_i c_i = 0$, so $\sum_i c_i \bar{y}_{i\cdot} = \sum_i c_i \tau_i + \sum_i c_i \bar{\varepsilon}_{i\cdot}$ exactly as with fixed blocks. The block means are $\bar{y}_{\cdot j} = \mu + \bar{\tau} + \beta_j + \bar{\varepsilon}_{\cdot j}$, and the $\beta_j + \bar{\varepsilon}_{\cdot j}$ are independent with variance $\sigma_\beta^2 + \sigma^2/t$, so $E(\text{SS}_B) = t(b-1)(\sigma_\beta^2 + \sigma^2/t)$ and $E(\text{MS}_B) = \sigma^2 + t\sigma_\beta^2$. The error sum of squares is unchanged, because the residual projection annihilates every vector that is constant within blocks, so $E(\text{MSE}) = \sigma^2$. Then

$$E\left(\frac{\text{MS}_B + (t-1)\text{MSE}}{tb}\right) = \frac{t\sigma^2 + t\sigma_\beta^2}{tb} = \frac{\sigma^2 + \sigma_\beta^2}{b}.$$

A comparison of treatments uses only within-block information, in which $\bar{\beta}$ cancels, so $s/\sqrt{b}$ is its scale and the Tukey intervals of Exercise (B1) remain exact. A single treatment mean is a statement about the population of blocks, and its uncertainty includes the sampling of the b blocks: s^2/b misses the term σ_β^2/b . The estimate above mixes two mean squares with different degrees of freedom, so an interval built on it is only approximately t ; the proper treatment is the subject of Chapter 32.